Deirdre Kimberley

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science - Mechanical Engineering — *Sept. 2022 – Present*

Experience

Apple — April 2026 – Aug. 2026
Manufacturing Quality Engineer – Home/Audio — *Cupertino, CA*

- Led the R&D of metrology systems for **previously unverifiable surface and optical quality characteristics**, from concept through full-scale equipment deployment
- Owned the DFM and dimensional convergence for 3 parts through early builds, closing **160+ nonconformances** through tolerance studies, Cpk/FAI validation, DOE trials, and datum/GD&T scheme convergence
- Delivered **2 production equipment programs on schedule**: a DFM-driven machine retrofit and a new production line quality control plan, aligning vendors and engineering teams from concept through implementation

Tesla — Sept. 2025 – Dec. 2025
Manufacturing Technical Program Manager – Optimus — *Palo Alto, CA*

- Owned **end-to-end manufacturing for the Optimus hand line**, coordinating cross-functional teams and defining 120+ manufacturing interfaces for automation equipment design
- Resolved **200+ build issues**, coordinating design, manufacturing, quality, and production teams to maintain build targets and drive DFM/DFA improvements
- Delivered **34 line fixtures and 9 testers** for the next generation hand, validating early designs through rapid prototyping and reliability testing

Tesla — Jan. 2025 – April 2025
Manufacturing Quality Engineer – Cybercab — *Austin, TX*

- Developed the wheel alignment PFMEA, identifying **100+ manufacturing risks** and implementing controls for the unboxed alignment process
- Led the R&D and production validation of an automated vision inspection system, developing test methodology and qualifying detection performance for line deployment
- Redesigned the manufacturing traceability process, identifying and driving closure of **200+ gaps** to mitigate line-down risk ahead of launch

Magna International — April 2024 – Aug. 2024
Research and Development Engineer — *Toronto, ON*

- Designed and implemented a cobot-based quality inspection station, developing rapid-deployment fixtures and operator software that reduced **setup time by 40%**
- Performed mechanical property testing and analyzed data across **2,000+ samples** to support R&D programs

Martinrea International — Sept. 2023 – Dec. 2023
Manufacturing Engineer — *Tillsonburg, ON*

- Designed and launched a consolidated manufacturing station, delivering **$225,000 in annual savings** while eliminating 450 sq. ft. of floor space, and 1 operator
- Redesigned production line stations using Design for Six Sigma, **increasing throughput by 50%**, reducing cycle time by **17 seconds**, and improving operator safety

Projects

Pick and Place Robot | *Python, Arduino, Motor Control, Sensor Integration* — Jan. 2026 – April 2026

- Designed the robot's electrical architecture, integrating motors, sensors, control hardware, and power systems to enable automated pick-and-place operation
- Developed motion-control and sequencing software, integrating sensor feedback and validating performance across mechanical and electrical subsystems

Technical Skills

Mechanical: SOLIDWORKS, CATIA, NX, AutoCAD, FEA, GD&T, Root Cause Analysis, Tolerance Analysis, Jira
Electrical: Arduino, Raspberry Pi, I2C, Circuit Design, Sensor Integration, Motor Control, PCB Design
Software: Python, C++, JavaScript, MATLAB, Excel, HTML, TypeScript, Vercel, React